

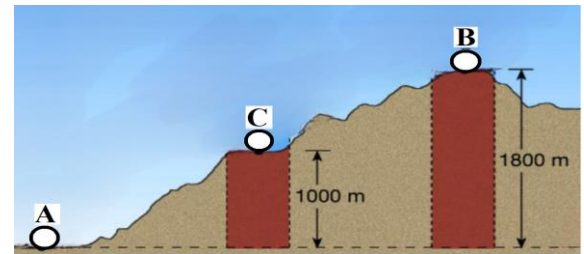
Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ MI/E	First Test - Modeule D	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **6 Marks**

1) A suck pressure is a negative gauge pressure		
2) A pressure does not change in Pascal principle,		
3) The electric flux is directly proportional to electric field		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) A Fully immersed object in fluid will sink if:				
a) $\rho_o < \rho_f$	b) $\rho_o > \rho_f$	c) $\rho_f > \rho_o$	d) $\rho_o = \rho_f$	e) None of these
2) 25 N object displaces 25 N of water. The Buoyant force is:				
a) 10 N	b) 25 N	c) 15 N	d) 0 N	e) None of these
3) From the figure shown below. The pressures				



a) $P_C > P_B$	b) $P_A < P_B$	c) $P_C < P_B$	d) $P_C = P_A$	e) None of these
4) The force 1 N. If a distance changes from 8 m into 2 m. The force is:				
a) 1 N	b) 0.5 N	c) 16 N	d) 128 N	e) None of these
5) The two spheres come into contact and then separate. The total charge on spheres A and B is:				



a) 1.5 C	b) 6 C	c) 0	d) 3 C	e) None of these
6) Which of the following has negative charge:				
a) Electron & Proton	b) Neutron	c) Proton	d) Electron	e) None of these
7) Object with 4 m×3m×2m having a density of x kg/m ³ . The pressure in Pa on the ground is:				
a) 20 x	b) 0.125 x	c) 2 x	d) 1.25 x	e) None of these

Section III: Answer all the following by selecting the correct choice
Full and clear steps are required **20 Marks**

Constant		
Electron: $m = 9.11 \times 10^{-31}$ kg	$K_e = 9 \times 10^9$ N.m ² /C ²	Gravity: $g = 10$ m/s ²
Proton: $m = 1.67 \times 10^{-27}$ kg	$G = 6.67 \times 10^{-11}$ N.m ² /kg ²	Water: $\rho_f = 1000$ kg/m ³

1) Two objects having $m_1 = 2$ kg and $m_2 = 2$ kg, separated by 300 cm. The gravitational force is:

a) 88.9×10^{-11} N	b) 44.47×10^{-11} N	c) 1.48×10^{-11} N	d) 2.96×10^{-11} N	e) None of these
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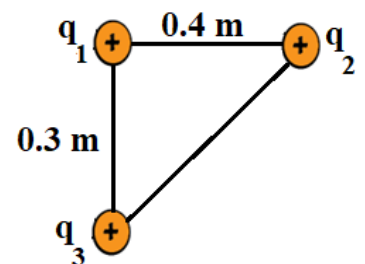
2) Two opposite charges one is 4 times the other separated by distance r , if a ratio of charge to a distance r is $2 \mu\text{C}/\text{m}$. The electric force on $3q$ placed at distance r from a point where $E = 0$ is:

a) 0.033 N	b) 3.6 N	c) 0 N	d) 72 N	e) None of these
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3) A room has dimensions of 3 m and 2 m and a height of 3 m. If the air density is $1.21 \text{ kg}/\text{m}^3$. The weight of the air in the room is:

a) 217.8 N	b) 290.4 N	c) 320.7 N	d) 435.6 N	e) None of these
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2) The total electric force on q_3 is: $q_1 = 2 \mu\text{C}$, $q_2 = 2 \mu\text{C}$ and $q_3 = 3 \mu\text{C}$



a) 0.46 N	b) 0.75 N	c) 0.37 N	d) 0.67 N	e) None of these
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5) 120 N object in air displaces 50 N when it is fully immersed in a fluid of $\rho_f = 1500 \text{ kg}/\text{m}^3$. The volume of the metal is:

a) 0.023 m^3	b) 0.0047 m^3	c) 0.008 m^3	d) 0.003 m^3	e) None of these
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